Chicago Neighborhood Explorer

Final Project Report

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Project Idea, Motivation, and System Requirements

Project Idea

Chicago Neighborhood Explorer is a database-driven application that helps users compare Chicago's 77 community areas through multiple dimensions of neighborhood life: demographics and housing affordability, city-supported affordable rental developments, public safety, CTA rail access and ridership, 311 service response, and ward/community-area geographic overlap.

The project integrates public civic datasets that are normally separated across different sources into one structured relational database. By using `CommunityArea` as the primary analytical geography, the application allows users to examine how neighborhood conditions differ across Chicago and to connect those conditions with relevant City Council ward geographies.

Motivation

Choosing or evaluating a neighborhood often requires considering several factors at the same time. A neighborhood with relatively affordable housing may differ in crime activity, rail access, service-response performance, or political reporting geography. Without an integrated database, users must search across disconnected datasets and manually compare results.

This project addresses that problem by building a unified analytical system that supports meaningful cross-dataset queries and presents results through an interactive interface.

Functional Requirements

The system must:

1. Store and connect official neighborhood-level data on demographics, affordability, affordable housing developments, crime incidents, CTA rail stations and ridership, 311 service requests, and ward overlaps.
2. Support comparison of community areas using safety, housing, transit, and service-response indicators.
3. Allow users to explore crime trends, CTA rail access and ridership, affordable housing near rail stations, and community areas overlapping selected wards.
4. Calculate derived analytical measures, such as housing-to-station distance and neighborhood profile indicators, through database queries rather than storing unnecessary redundant relationships.
5. Provide a graphical user interface in which users select predefined queries, view readable results, and insert a new local service-request record into the project database.

Implementation Overview

The final implementation uses MySQL 8.0 as the backend relational database, Python for data preprocessing and database loading, and Streamlit for the graphical user interface. Activity-based analysis is based on full calendar year 2025 data, while housing and rail-station measures represent imported infrastructure snapshots.

The relational approach is appropriate because the project depends on structured entities, primary and foreign keys, many-to-many bridge relationships, joins across official datasets, and aggregation queries for neighborhood comparison.

Final Conceptual ER Diagram and Key Revision

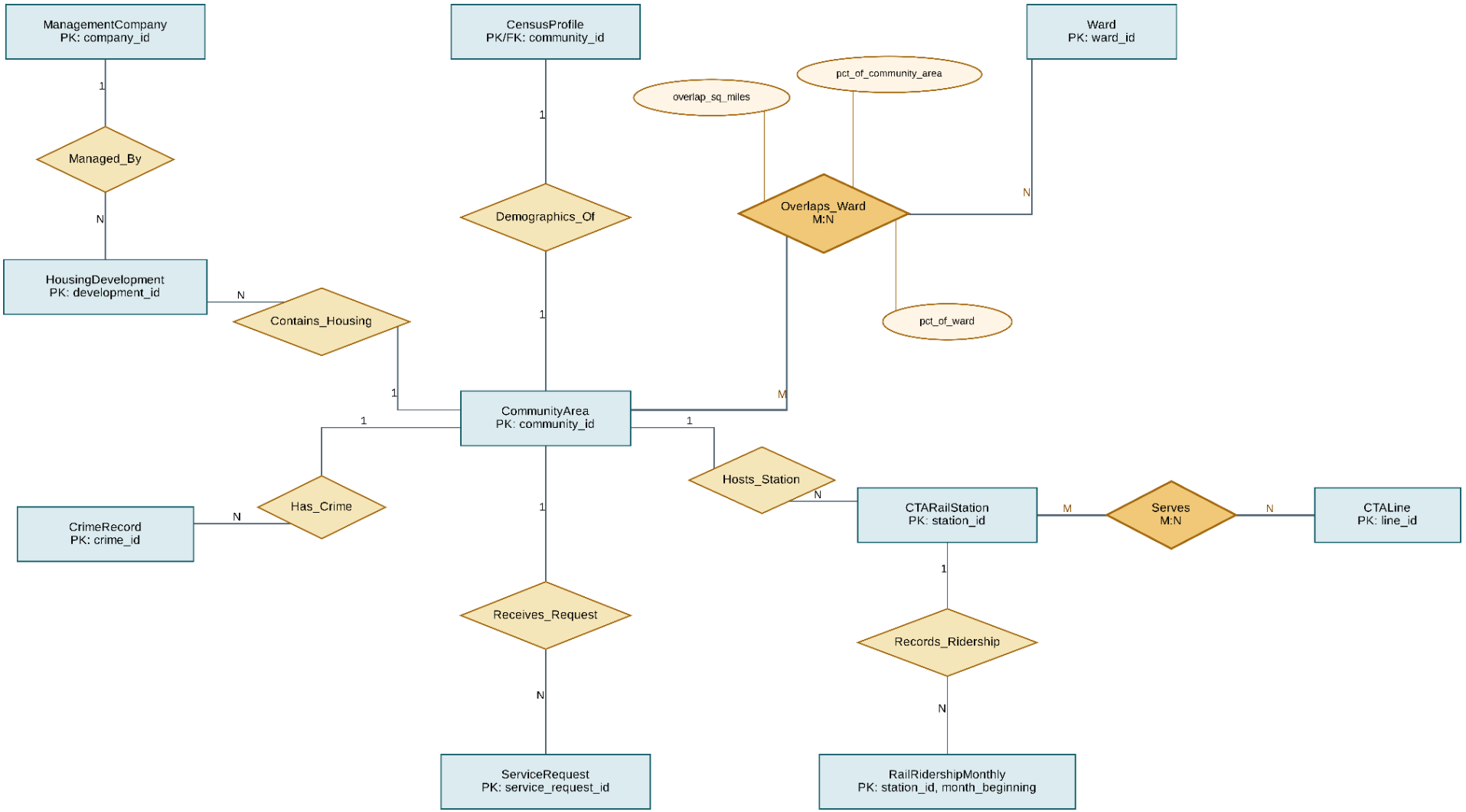


Figure 1. Final conceptual ER diagram. The highlighted M:N relationships are Overlaps_Ward and Serves.

Design adjustment. In response to instructor feedback, the former Near_Stop relationship was removed: housing-to-station distance is calculated dynamically from stored coordinates rather than persisted as a relationship. The final second M:N relationship is Overlaps_Ward, materialized in MySQL as community_area_ward_overlap. The loaded bridge contains 258 positive-area overlap rows; validation confirms M:N multiplicity because community areas 24 and 28 each overlap seven wards, while Ward 8 overlaps ten community areas.

Final Relational Schema Diagram

Teal boxes are base tables. Gold boxes are the two materialized bridge tables implementing the required M:N relationships. Gray boxes are derived SQL views, not base tables.

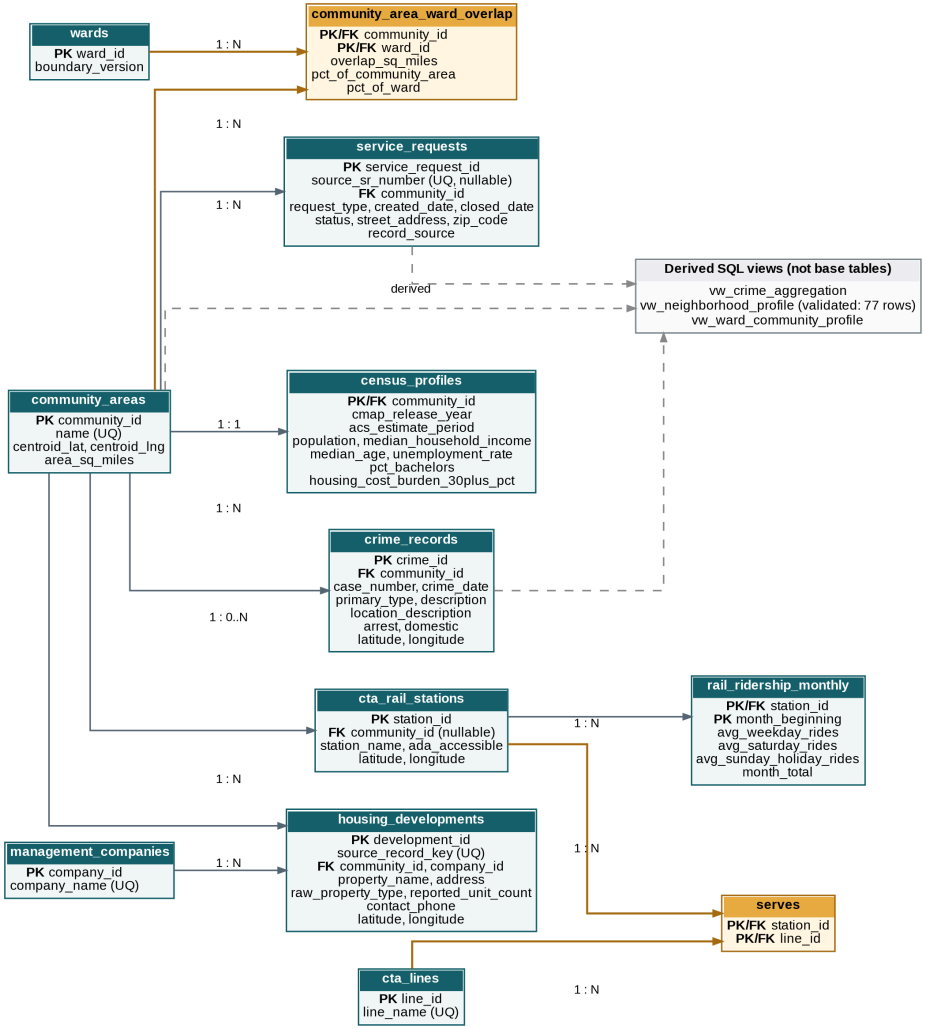


Figure 2. Final relational schema representation for the MySQL implementation.

Schema adjustment summary. The final implementation contains 12 base tables. Near_Stop, HousingCategory, DevelopmentCategory, crime_aggregations, and neighborhood_profiles are not base tables. Derived analytical results are represented through SQL views; vw_neighborhood_profile has been validated to return one row for each of the 77 community areas.

Data Sources and ETL Scope

The database is populated from official Chicago and CMAP datasets and deterministic transformations of those datasets. Chicago's 77 community areas are the analytical anchor. Activity measures use full calendar year 2025 records; housing and CTA station measures are imported snapshots. No randomly generated analytical records are used.

Official source / derivation	Relation(s) populated	Attributes covered	Loaded rows / preprocessing evidence
City of Chicago Community Area Boundaries (igwz-8jzy)	community_areas	Community_id (1-77); name; centroid lat/lng; area in sq miles	77 areas
City of Chicago Ward Boundaries 2023-present (p293-wvbd)	Wards; community_area_ward_overlap	Ward_id (1-50); boundary_version; GIS-intersected overlap_sq_miles; overlap area and overlap percentages	50 wards; 258 bridge rows; M:N validated
CMAP Community Data Snapshots / ACS estimates	census_profiles	Population; household income; age; unemployment; bachelor-degree; housing cost burden	77 rows; one per community area
Affordable Rental Housing Developments (s6ha-ppgi)	housing_developments; management_companies	Property; address; type; reported units; contact/company; coordinates	598 raw; 596 developments after duplicate removal; 236 companies
Crimes - 2001 to Present (ijzp-q8t2), 2025 only	crime_records	Official crime ID; datetime; type; description; flags; location; community	237,274 records; deduplicated only by official crime ID
CTA List of ‘L’ Stops (8pix-ypme)	cta_rail_stations; cta_lines; serves	Station/MAP_ID; coordinate; ADA; rail-line membership	144 stations; 8 lines; 191 links; 20 outside Chicago scope
CTA ‘L’ Monthly Day-Type Averages and Totals (t2rn-p8d7), 2025 only	rail_ridership_monthly	Station-month; weekday/Sat/Sun-holiday averages; monthly total	1,728 station-month rows; no duplicate composite keys
311 Service Requests (v6vf-nfxy), 2025 only	service_requests	SR number; type; created/closed datetime; status; location; source	1,885,931 official OPEN_DATA rows
SQL derivation from loaded relations	vw_neighborhood_profile; other analytical views	Neighborhood metrics; ward crosswalk display; monthly crime summaries	vw_neighborhood_profile validated: 77 rows

Source note: detailed ETL transformations, date-range checks, and screenshots are included in Appendix A and Appendix B of this component draft. Full citations listed in Appendix C.3.

SQL Queries and Output Screenshots

```sql
SELECT
    w.ward_id,
    ca.name AS community_name,
    ROUND(o.overlap_sq_miles, 4) AS overlap_sq_miles,
    ROUND(o.pct_of_ward, 2) AS pct_of_ward,
    ROUND(o.pct_of_community_area, 2) AS pct_of_community_area,
    p.crime_incidents_per_1000_population_estimate_2025 AS crime_rate,
    p.housing_cost_burden_30plus_pct AS cost_burden_pct,
    p.avg_closed_311_response_hours_2025 AS avg_311_hours
FROM wards w
JOIN community_area_ward_overlap o
    ON w.ward_id = o.ward_id
JOIN community_areas ca
    ON o.community_id = ca.community_id
LEFT JOIN vw_neighborhood_profile p
    ON ca.community_id = p.community_id
WHERE w.ward_id = 8
ORDER BY o.pct of ward DESC, ca.name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ward_id	community_name	overlap_sq_miles	pct_of_ward	pct_of_community_area	crime_rate	cost_burden_pct	avg_311_hours
8	AVALON PARK	1.2502	19.36	100.00	111.25	39.620	596.18
8	PULLMAN	1.0235	15.85	48.36	162.29	36.768	500.65
8	CALUMET HEIGHTS	0.9204	14.26	52.55	114.47	33.519	577.49
8	CHATHAM	0.8972	13.90	30.38	174.23	48.780	447.32
8	SOUTH SHORE	0.6982	10.82	23.79	149.21	51.815	519.25
8	BURNSIDE	0.6096	9.44	100.00	117.32	26.975	611.71
8	ROSELAND	0.3900	6.04	8.10	144.52	42.437	491.42
8	SOUTH CHICAGO	0.3599	5.58	10.76	126.75	44.328	539.78
8	GREATER GRAND CROSSING	0.2618	4.06	7.38	206.44	50.231	394.46
8	SOUTH DEERING	0.0454	0.70	0.42	130.12	41.921	428.12

Result 1

Figure 1. Query output for community areas overlapping Ward 8. The query returns ten community areas associated with Ward 8 through the `community_area_ward_overlap` bridge table and reports their overlap percentages and selected neighborhood indicators. This result demonstrates the ward-to-many-community direction of the final schema's second substantive many-to-many relationship.

Appendix A. Data Pipeline Method and Validation Evidence

A.1 ETL Implementation Scope

Python is used to download or read source extracts, clean and transform records, compute spatial overlays, load MySQL-ready data, and validate the resulting database. The data pipeline produces official-data-backed base relations and SQL views for downstream Streamlit queries.

Pipeline decision	Final implementation
Backend DBMS	MySQL 8.0
ETL language	Python; GeoPandas/Shapely for spatial transformations
GUI integration	Streamlit queries MySQL in real time
Geographic anchor	Chicago's 77 community areas
Activity analysis window	January 1 through December 31, 2025
Stored M:N relationships	serves; community_area_ward_overlap
Removed derived relationship	Near_Stop; distance-to-rail calculated at query time

A.2 Core Cleaning and Transformation Rules

ETL area	Final rule
Community areas and wards	Project official polygons from community areas and wards to Illinois State Plane; compute community area/centroid values; spatially intersect area and ward polygons to build the bridge table.
Ward overlap bridge	Keep positive-area intersections after the final sliver-filtering rule; store overlap_sq_miles, pct_of_community_area, and pct_of_ward.
CMAP profile data	Map official abbreviated fields into demographic and housing-cost-burden measures; retain one profile row per community area.
Housing	Generate SHA-256 source_record_key; remove two duplicated source rows; retain phone at development-record level.
CTA station scope	Deduplicate stop records to MAP_ID station level; use polygon covers mapping; keep outside-Chicago stations with NULL community_id and exclude them from neighborhood aggregation.
Crime	Filter to calendar year 2025; use official crime ID as the primary key; do not deduplicate by case number.
Rail ridership	Load official station-month data without unpivoting away month_total; validate uniqueness of (station_id, month_beginning).
311	Filter official records to 2025; load with record_source='OPEN_DATA'; reserve 'GUI_INPUT' for local app insertions excluded from official response metrics.

A.3 Loaded Relation and View Counts

Relation / view	Rows	Relation / view	Rows
community_areas	77	wards	50
community_area_ward_overlap	258	census_profiles	77

Relation / view	Rows	Relation / view	Rows
management_companies	236	housing_developments	596
crime_records	237,274	cta_rail_stations	144
cta_lines	8	serves	191
rail_ridership_monthly	1,728	service_requests (OPEN_DATA)	1,885,931
vw_neighborhood_profile	77		

A.4 Activity-Date Validation

Relation	Minimum date	Maximum date	Rows
crime_records	2025-01-01 00:00:00	2025-12-31 23:58:00	237,274
rail_ridership_monthly	2025-01-01	2025-12-01	1,728
service_requests where record_source='OPEN_DATA'	2025-01-01 00:01:27	2025-12-31 23:59:32	1,885,931

Rail ridership is stored at the station-month grain; therefore, the December 2025 period is represented by month_beginning = '2025-12-01'.

A.5 M:N Relationship Validation Evidence

The final bridge table is not a threshold-based proximity table. It is a crosswalk created from two official, versioned polygon geography layers. The validation output provides witnesses in both directions of the relationship.

Community areas overlapping multiple wards

community_id	ward_count
24	7
28	7
15	6
19	6
30	6
61	6
66	6
71	6
6	5
16	5

Wards overlapping multiple community areas

ward_id	community_count
8	10
3	9
20	9
28	9
39	8
4	7
9	7
10	7
11	7
14	7

Interpretation. Since individual community areas intersect multiple wards and individual wards intersect multiple community areas, CommunityArea and Ward form a genuine M:N relationship implemented by community_area_ward_overlap.

A.6 Transit and Data-Integrity Checks

Validation check	Result	Interpretation
CTA stations in master table	144	Full station reference list retained.
Stations outside Chicago community-area scope	20	community_id is NULL; excluded from neighborhood-level measures.
Chicago-mapped stations used in neighborhood analysis	124	144 total minus 20 outside scope.
Duplicate rail station-month keys	0 rows	Composite-key grain is valid.
Service request sources at load time	OPEN_DATA: 1,885,931	GUI_INPUT: 1 (verified via Streamlit insertion test)

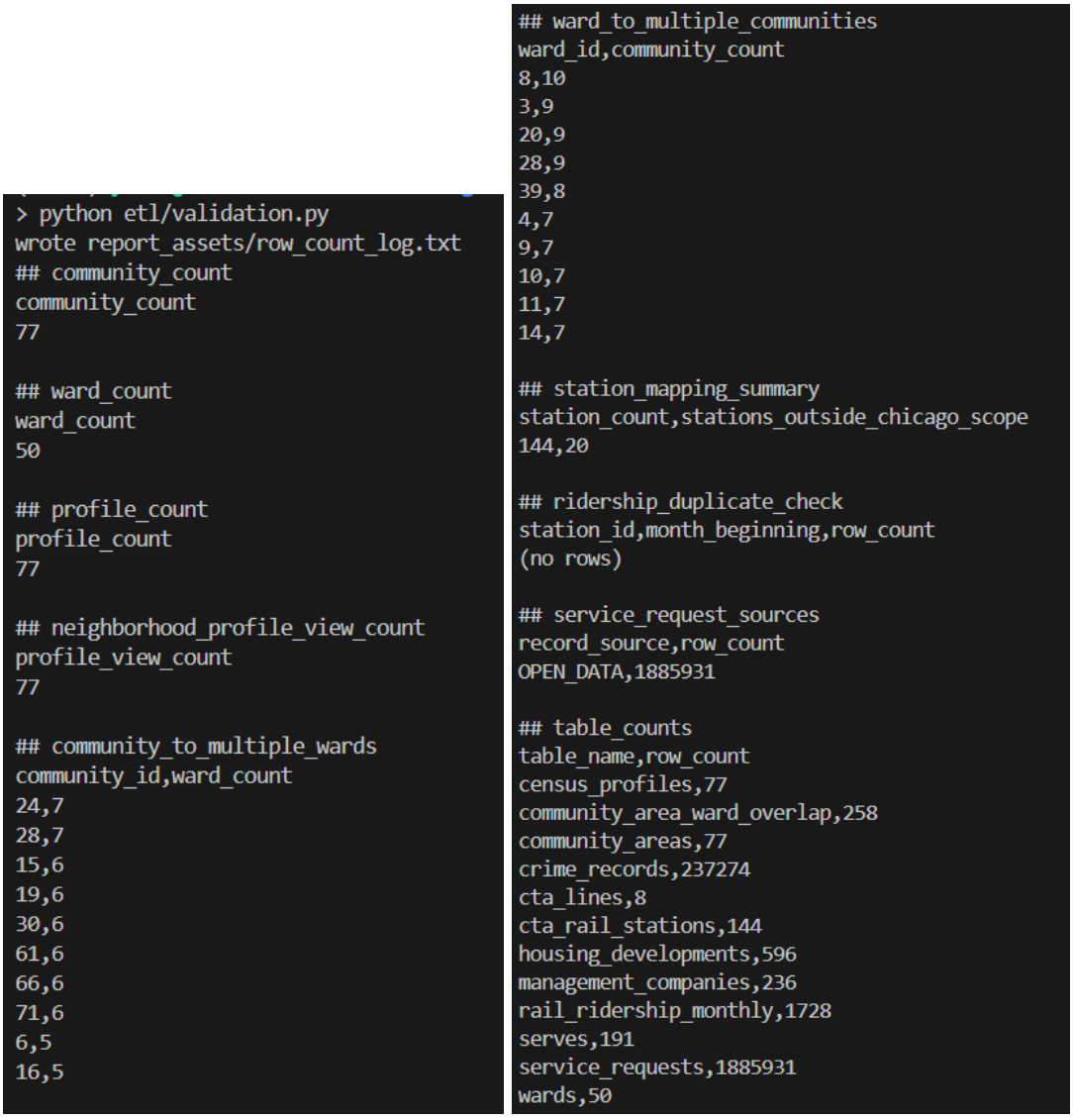
Appendix B. Screenshot Placement Sheet

```
python data/prepare_data.py \
  --skip-download \
  --processed-dir report_assets/b1_processed_demo \
  2>&1 | tee report_assets/B1_prepare_data_cached_raw.log
INFO: using existing raw file data/raw/community_areas.geojson
INFO: wrote report_assets/b1_processed_demo/community_areas.csv (77 rows)
INFO: using existing raw file data/raw/wards_2023.geojson
INFO: wrote report_assets/b1_processed_demo/wards.csv (50 rows)
INFO: wrote report_assets/b1_processed_demo/community_area_ward_overlap.csv (258 rows)
INFO: wrote report_assets/b1_processed_demo/census_profiles.csv (77 rows)
INFO: using existing raw file data/raw/affordable_rental_housing_developments.csv
INFO: wrote report_assets/b1_processed_demo/management_companies.csv (236 rows)
INFO: wrote report_assets/b1_processed_demo/housing_developments.csv (596 rows)
INFO: using existing raw file data/raw/cta_l_stops.csv
INFO: wrote report_assets/b1_processed_demo/cta_rail_stations.csv (144 rows)
INFO: wrote report_assets/b1_processed_demo/cta_lines.csv (8 rows)
INFO: wrote report_assets/b1_processed_demo/serves.csv (191 rows)
INFO: using existing raw file data/raw/cta_rail_ridership_monthly_2025.csv
INFO: wrote report_assets/b1_processed_demo/rail_ridership_monthly.csv (1728 rows)
INFO: using existing raw file data/raw/crimes_2025.csv
INFO: wrote report_assets/b1_processed_demo/crime_records.csv (237274 rows)
INFO: using existing raw file data/raw/service_requests_2025_01.csv
INFO: using existing raw file data/raw/service_requests_2025_02.csv
INFO: using existing raw file data/raw/service_requests_2025_03.csv
INFO: using existing raw file data/raw/service_requests_2025_04.csv
INFO: using existing raw file data/raw/service_requests_2025_05.csv
INFO: using existing raw file data/raw/service_requests_2025_06.csv
INFO: using existing raw file data/raw/service_requests_2025_07.csv
INFO: using existing raw file data/raw/service_requests_2025_08.csv
INFO: using existing raw file data/raw/service_requests_2025_09.csv
INFO: using existing raw file data/raw/service_requests_2025_10.csv
INFO: using existing raw file data/raw/service_requests_2025_11.csv
INFO: using existing raw file data/raw/service_requests_2025_12.csv
INFO: wrote report_assets/b1_processed_demo/service_requests.csv
INFO: all processed CSV files are ready in report_assets/b1_processed_demo
```

Figure B1. Rerun of the Python preprocessing pipeline using cached raw source extracts. The script reuses previously downloaded raw files and writes regenerated MySQL-ready CSV outputs to a temporary report-assets directory; it does not modify the populated MySQL database.

```
> python -u etl/load_database.py \
  --create-schema \
  --create-indexes \
  2>&1 | tee report_assets/B2_clean_database_rebuild.log
loaded community_areas: 77 rows
loaded wards: 50 rows
loaded community_area_ward_overlap: 258 rows
loaded census_profiles: 77 rows
loaded management_companies: 236 rows
loaded housing_developments: 596 rows
loaded crime_records: 237274 rows
loaded cta_rail_stations: 144 rows
loaded cta_lines: 8 rows
loaded serves: 191 rows
loaded rail_ridership_monthly: 1728 rows
loaded service_requests: 1885931 rows
community_areas: 77
wards: 50
community_area_ward_overlap: 258
census_profiles: 77
management_companies: 236
housing_developments: 596
crime_records: 237274
cta_rail_stations: 144
cta_lines: 8
serves: 191
rail_ridership_monthly: 1728
service_requests: 1885931
```

Figure B2. Python-to-MySQL connectivity and post-load verification. The loader application connects to the populated MySQL database and reports loaded relation counts in verification-only mode without recreating tables, reloading data, or modifying derived views.



```
> python etl/validation.py
wrote report_assets/row_count_log.txt
## community_count
community_count
77

## ward_count
ward_count
50

## profile_count
profile_count
77

## neighborhood_profile_view_count
profile_view_count
77

## community_to_multiple_wards
community_id,ward_count
24,7
28,7
15,6
19,6
30,6
61,6
66,6
71,6
6,5
16,5

## ward_to_multiple_communities
ward_id,community_count
8,10
3,9
20,9
28,9
39,8
4,7
9,7
10,7
11,7
14,7

## station_mapping_summary
station_count,stations_outside_chicago_scope
144,20

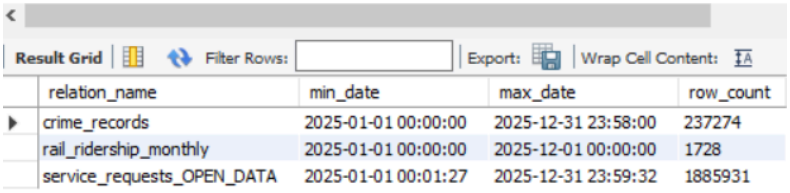
## ridership_duplicate_check
station_id,month_beginning,row_count
(no rows)

## service_request_sources
record_source,row_count
OPEN_DATA,1885931

## table_counts
table_name,row_count
census_profiles,77
community_area_ward_overlap,258
community_areas,77
crime_records,237274
cta_lines,8
cta_rail_stations,144
housing_developments,596
management_companies,236
rail_ridership_monthly,1728
serves,191
service_requests,1885931
wards,50
```

Figure B3. Post-load validation of final database relations and analytical view. The validation results confirm that the final MySQL database contains all 77 Chicago community areas, 50 wards, 258 community-area/ward overlap bridge records, and one derived neighborhood profile row for each community area. The database also includes substantial real-world activity data, including 237,274 crime records and 1,885,931 official 311 service-request records.

```sql
SELECT
    'crime_records' AS relation_name,
    MIN(crime_date) AS min_date,
    MAX(crime_date) AS max_date,
    COUNT(*) AS row_count
FROM crime_records
UNION ALL
SELECT
    'rail_ridership_monthly',
    MIN(month_beginning),
    MAX(month_beginning),
    COUNT(*)
FROM rail_ridership_monthly
UNION ALL
SELECT
    'service_requests_OPEN_DATA',
    MIN(created_date),
    MAX(created_date),
    COUNT(*)
FROM service_requests
WHERE record_source = 'OPEN_DATA';
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

relation_name	min_date	max_date	row_count
crime_records	2025-01-01 00:00:00	2025-12-31 23:58:00	237274
rail_ridership_monthly	2025-01-01 00:00:00	2025-12-01 00:00:00	1728
service_requests_OPEN_DATA	2025-01-01 00:01:27	2025-12-31 23:59:32	1885931

Figure B4. Full-calendar-year 2025 activity-data validation. The loaded activity tables span the intended 2025 analysis window. The `crime_records` table contains 237,274 incidents dated from January 1 through December 31, 2025, while `service_requests` contains 1,885,931 official open-data requests created during the same calendar-year period. CTA rail ridership is stored at monthly grain and includes 1,728 station-month records from January through December 2025, represented by `month_beginning` values from `2025-01-01` through `2025-12-01`.

```sql
SELECT
    ward_id,
    community_name,
    ROUND(overlap_sq_miles, 4) AS overlap_sq_miles,
    ROUND(pct_of_ward, 2) AS pct_of_ward,
    ROUND(pct_of_community_area, 2) AS pct_of_community_area,
    crime_incidents_per_1000_population_estimate_2025 AS crime_rate,
    housing_cost_burden_30plus_pct AS cost_burden_pct,
    avg_closed_311_response_hours_2025 AS avg_311_hours
FROM vw_ward_community_profile
WHERE ward_id = 8
ORDER BY pct_of_ward DESC, community_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ward_id	community_name	overlap_sq_miles	pct_of_ward	pct_of_community_area	crime_rate	cost_burden_pct	avg_311_hours
8	AVALON PARK	1.2502	19.36	100.00	111.25	39.620	596.18
8	PULLMAN	1.0235	15.85	48.36	162.29	36.768	500.65
8	CALUMET HEIGHTS	0.9204	14.26	52.55	114.47	33.519	577.49
8	CHATHAM	0.8972	13.90	30.38	174.23	48.780	447.32
8	SOUTH SHORE	0.6982	10.82	23.79	149.21	51.815	519.25
8	BURNSIDE	0.6096	9.44	100.00	117.32	26.975	611.71
8	ROSELAND	0.3900	6.04	8.10	144.52	42.437	491.42
8	SOUTH CHICAGO	0.3599	5.58	10.76	126.75	44.328	539.78
8	GREATER GRAND CROSSING	0.2618	4.06	7.38	206.44	50.231	394.46
8	SOUTH DEERING	0.0454	0.70	0.42	130.12	41.921	428.12

Figure B5. Analytical query using the `CommunityAreaWardOverlap` many-to-many relationship. This query retrieves the community areas overlapping Ward 8 and displays their overlap measures together with neighborhood-level crime, housing cost-burden, and 311 response-time indicators. The result returns ten overlapping community areas, demonstrating the practical use of the ward/community-area bridge table in the final application.

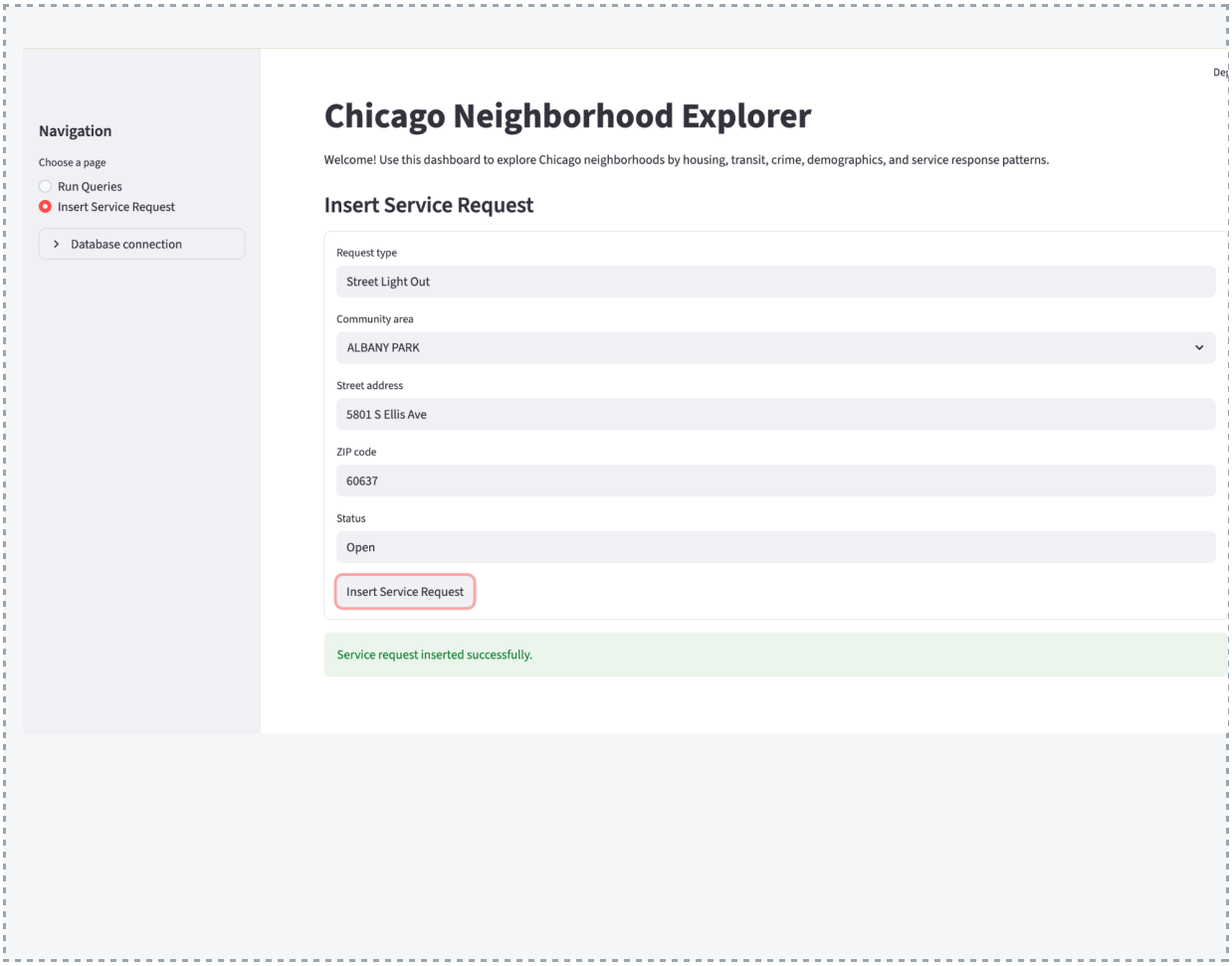


Figure B6. Verification of GUI-based insertion into the `service_requests` table. After one service-request record is added through the Streamlit interface, the database distinguishes the locally entered `GUI_INPUT` row from the 1,885,931 imported `OPEN_DATA` rows. This design verifies that the application supports record insertion while preventing demonstration or test inputs from affecting official neighborhood-level analytical metrics.

Appendix C. SQL Evidence Templates and Source Registry

C.1 Required Ward Query Screenshot Template

This query demonstrates the ward/community-area M:N bridge in actual application use.

```
SELECT
    ward_id,
    community_name,
    ROUND(overlap_sq_miles, 4) AS overlap_sq_miles,
    ROUND(pct_of_ward, 2) AS pct_of_ward,
    ROUND(pct_of_community_area, 2) AS pct_of_community_area,
    crime_incidents_per_1000_population_estimate_2025 AS crime_rate,
    housing_cost_burden_30plus_pct AS cost_burden_pct,
    avg_closed_311_response_hours_2025 AS avg_311_hours
FROM vw_ward_community_profile
WHERE ward_id = 8
ORDER BY pct_of_ward DESC, community_name;
```

C.2 Screenshot-Supporting Validation Queries

```
SELECT COUNT(*) AS community_count FROM community_areas;
SELECT COUNT(*) AS ward_count FROM wards;
SELECT COUNT(*) AS overlap_rows FROM community_area_ward_overlap;
SELECT COUNT(*) AS profile_rows FROM vw_neighborhood_profile;

SELECT community_id, COUNT(DISTINCT ward_id) AS ward_count
FROM community_area_ward_overlap
GROUP BY community_id HAVING COUNT(DISTINCT ward_id) > 1
ORDER BY ward_count DESC LIMIT 10;

SELECT ward_id, COUNT(DISTINCT community_id) AS community_count
FROM community_area_ward_overlap
GROUP BY ward_id HAVING COUNT(DISTINCT community_id) > 1
ORDER BY community_count DESC LIMIT 10;
```

C.3 Source Registry

ID	Reference / use in final report
[C1]	MPCS 53001 Databases. Final Project - Step 2 assignment handout. Requires at least eight entity sets, six relationship sets, and two M:N relationships.
[C2]	MPCS 53001 Databases. Final Project - Step 3 assignment handout. Requires final ER/schema/data-source pages and SQL output screenshots, and permits implementation-driven schema revisions.
[D1]	City of Chicago Data Portal. Boundaries - Community Areas (dataset ID igwz-8jzy).
[D2]	City of Chicago Data Portal. Boundaries - Wards (2023-) (dataset ID p293-wvbd).
[D3]	City of Chicago Data Portal. Affordable Rental Housing Developments (dataset ID s6ha-ppgi).
[D4]	City of Chicago Data Portal. CTA - System Information - List of ‘L’ Stops (dataset ID 8pix-ypme).
[D5]	City of Chicago Data Portal. CTA - Ridership - ‘L’ Station Entries - Monthly Day-Type Averages & Totals (dataset ID t2rn-p8d7).
[D6]	City of Chicago Data Portal. Crimes - 2001 to Present (dataset ID ijzp-q8t2).
[D7]	City of Chicago Data Portal. 311 Service Requests (dataset ID v6vf-nfxy).

ID	Reference / use in final report
[D8]	Chicago Metropolitan Agency for Planning (CMAP). Community Data Snapshots / ACS estimates.
[V1]	Data Pipeline Lead validation log, May 2026. Source of final loaded row counts, date-range checks, M:N multiplicity witnesses, and transit-scope checks.

Appendix D. SQL/output section

```
QUERY_OPTIONS = {
    "q1_affordable_safe": {
        "label": "Q1 Affordable + Safe",
        "description": "Find neighborhoods with affordable housing and lower safety risk.",
    },
    "q2_housing_near_transit": {
        "label": "Q2 Housing Near Transit",
        "description": "Find neighborhoods with housing options near CTA access.",
    },
    "q3_transit_usage": {
        "label": "Q3 Transit Usage",
        "description": "Compare transit usage by selected year.",
    },
    "q5_high_demand_efficient": {
        "label": "Q5 High Demand + Efficient",
        "description": "Find areas with strong demand and efficient service response.",
    },
    "q6_most_accessible": {
        "label": "Q6 Most Accessible",
        "description": "Rank the most transit-accessible neighborhoods.",
    },
    "q7_crime_near_housing": {
        "label": "Q7 Crime Near Housing",
        "description": "Compare crime patterns near affordable housing.",
    },
    "q8_transit_popularity": {
        "label": "Q8 Transit Popularity",
        "description": "Rank neighborhoods by transit popularity.",
    },
    "q9_service_delays": {
        "label": "Q9 Service Delays",
        "description": "Find neighborhoods with the longest service delay patterns.",
    },
    "q10_demographics_vs_crime": {
        "label": "Q10 Demographics vs Crime",
        "description": "Compare demographic indicators with crime levels.",
    },
    "q11_neighborhood_ranking": {
        "label": "Q11 Neighborhood Ranking",
        "description": "Show an overall ranked list of neighborhoods.",
    },
    "q12_housing_availability": {
        "label": "Q12 Housing Availability",
        "description": "Summarize affordable housing availability.",
    },
    "q14_ward_overlap": {
        "label": "Q14 Ward Overlap",
        "description": "Show which community areas overlap with a selected ward.",
    },
}
```

```sql
USE chicago_neighborhood;

-- Query 1: Compare neighborhood profiles.
SELECT
    community_name,
    crime_incidents_per_1000_population_estimate_2025 AS crime_rate,
    housing_cost_burden_30plus_pct AS cost_burden_pct,
    reported_city_supported_units_snapshot AS reported_units,
    rail_station_density_snapshot AS station_density,
    avg_closed_311_response_hours_2025 AS avg_311_hours
FROM vw_neighborhood_profile
ORDER BY crime_rate ASC, avg_311_hours ASC
LIMIT 10;

-- Query 2: View crime trends for a selected community and optional crime type.
SET @community_id = 25;
SET @crime_type = NULL;

SELECT crime_month, crime_type, crime_count
FROM vw_crime_aggregation
WHERE community_id = @community_id
  AND crime_year = 2025
  AND (@crime_type IS NULL OR crime_type = @crime_type)
ORDER BY crime_month, crime_type;

-- Query 3: View community areas overlapping a selected ward.
SET @ward_id = 27;

SELECT
    ward_id,
    community_name,
    ROUND(overlap_sq_miles, 4) AS overlap_sq_miles,
    ROUND(pct_of_ward, 2) AS pct_of_ward,
    ROUND(pct_of_community_area, 2) AS pct_of_community_area,
    crime_incidents_per_1000_population_estimate_2025 AS crime_rate,
    housing_cost_burden_30plus_pct AS cost_burden_pct,
    avg_closed_311_response_hours_2025 AS avg_311_hours
FROM vw_ward_community_profile
WHERE ward_id = @ward_id
ORDER BY pct_of_ward DESC, community_name;
```

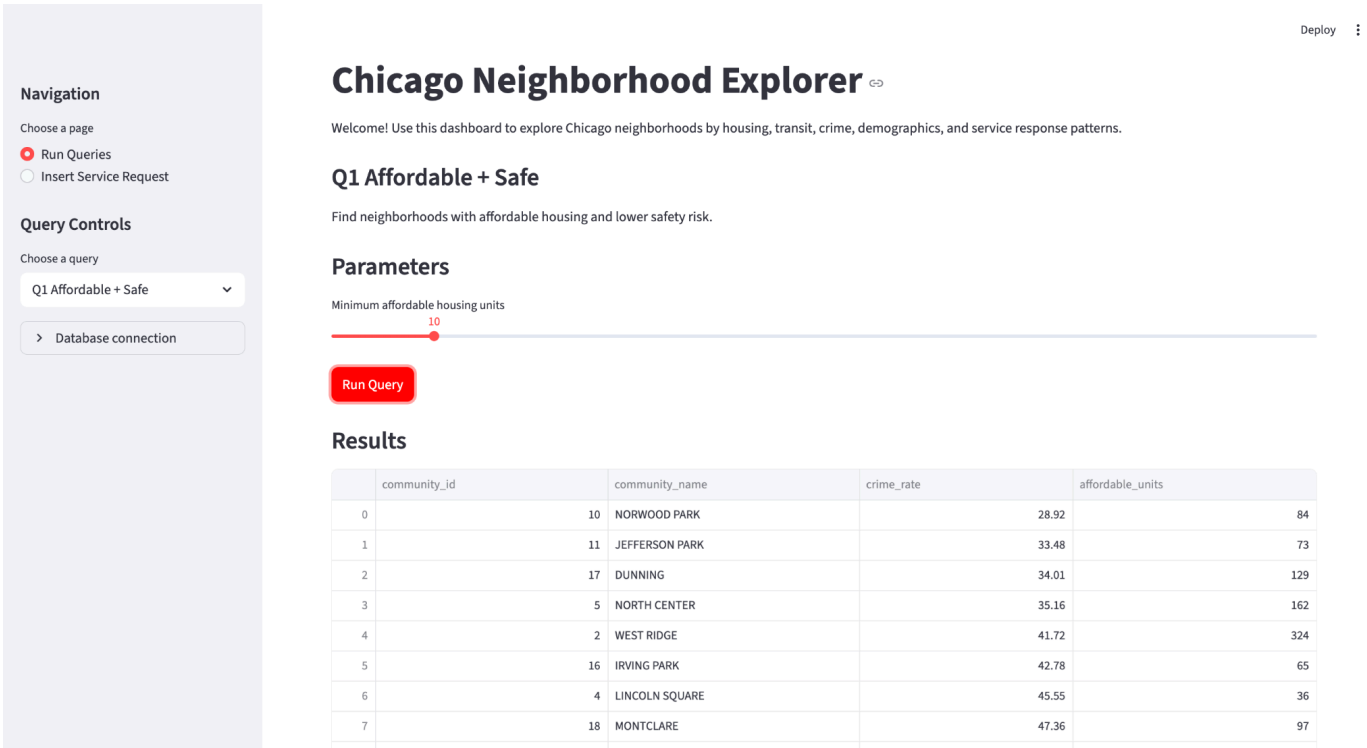
```sql
-- Query 4: Find city-supported affordable housing near CTA rail stations.
SET @community_id = 25;
SET @max_distance_meters = 800;

WITH candidate_distances AS (
    SELECT
        h.property_name,
        ca.name AS community_name,
        s.station_id,
        s.station_name,
        ST_Distance_Sphere(
            POINT(h.longitude, h.latitude),
            POINT(s.longitude, s.latitude)
        ) AS distance_meters
    FROM housing_developments h
    JOIN community_areas ca
        ON h.community_id = ca.community_id
    CROSS JOIN cta_rail_stations s
    WHERE h.community_id = @community_id
      AND h.latitude IS NOT NULL
      AND h.longitude IS NOT NULL
      AND s.latitude IS NOT NULL
      AND s.longitude IS NOT NULL
)
SELECT
    cd.property_name,
    cd.community_name,
    cd.station_name,
    l.line_name,
    ROUND(cd.distance_meters, 1) AS distance_meters
FROM candidate_distances cd
LEFT JOIN serves sv
    ON cd.station_id = sv.station_id
LEFT JOIN cta_lines l
    ON sv.line_id = l.line_id
WHERE cd.distance_meters <= @max_distance_meters
ORDER BY cd.distance_meters, cd.property_name, l.line_name;

-- Query 5: Display neighborhoods with the highest CTA rail entries in 2025.
SELECT community_name, rail_entries_2025, rail_station_count_snapshot
FROM vw_neighborhood_profile
ORDER BY rail_entries_2025 DESC
LIMIT 10;

-- Query 6: Compare average closed 311 response time in 2025.
SELECT community_name, avg_closed_311_response_hours_2025
FROM vw_neighborhood_profile
WHERE avg_closed_311_response_hours_2025 IS NOT NULL
ORDER BY avg_closed_311_response_hours_2025 ASC
LIMIT 10;
```

```sql
-- Insert feature: add one local project service-request record.
INSERT INTO service_requests (
    source_sr_number,
    request_type,
    created_date,
    closed_date,
    status,
    street_address,
    zip_code,
    community_id,
    record_source
) VALUES (
    NULL,
    'Street Light Out',
    NOW(),
    NULL,
    'Open',
    '5801 S Ellis Ave',
    '60637',
    41,
    'GUI_INPUT'
);
```



Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q2 Housing Near Transit

Database connection

Chicago Neighborhood Explorer

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Q2 Housing Near Transit

Find neighborhoods with housing options near CTA access.

Parameters

Minimum nearby transit stops

2

Run Query

Results

	development_id	property_name	property_type	total_units	address	community_area	num_stops
0	2	1000M	ARO	23	1000 S. Michigan Ave.	LOOP	16
1	299	Imprint Lofts	ARO	7	739 S. Clark St.	LOOP	16
2	583	Millie on Michigan	ARO	7	300 N. Michigan Ave.	LOOP	16
3	231	Randolph Tower City Apts. (st	Multifamily	16	188 W. Randolph St.	LOOP	16
4	287	1050 Van Buren	ARO	16	1050 W. Van Buren St.	NEAR WEST SIDE	10
5	289	1125 Van Buren LLC	ARO	5	1125 W. Van Buren St.	NEAR WEST SIDE	10
6	277	1218 W Lexington St	ARO	3	1218 W. Lexington St.	NEAR WEST SIDE	10
7	108	171 N. Aberdeen St.	ARO	6	171 N. Aberdeen St.	NEAR WEST SIDE	10

Deploy

Navigation

Choose a page

- Run Queries
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Query Controls

Choose a query

Q3 Transit Usage

Database connection

Chicago Neighborhood Explorer

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Q3 Transit Usage

Compare transit usage by selected year.

Parameters

Year

2025

Run Query

Results

	community_id	community_area	num_stations	total_entries	avg_weekday_rides
0	32	LOOP	16	26534554	5222
1	8	NEAR NORTH SIDE	7	11242982	4703
2	6	LAKE VIEW	7	8397809	3472
3	28	NEAR WEST SIDE	10	6454498	2076
4	24	WEST TOWN	4	4346697	3244
5	76	OHARE	2	4031400	5803
6	22	LOGAN SQUARE	3	3748343	3830

Deploy

Navigation

Choose a page

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- Insert Service Request

Query Controls

Choose a query

Q5 High Demand + Efficient

Database connection

Chicago Neighborhood Explorer

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Q5 High Demand + Efficient

Find areas with strong demand and efficient service response.

Parameters

Year

2025

Maximum average service hours

720

Run Query

Results

	community_id	community_area	total_requests	avg_response_hours
0	76	OHARE	367479	3.86
1	28	NEAR WEST SIDE	680239	13.06
2	24	WEST TOWN	31095	374.36
3	22	LOGAN SQUARE	24762	398.29
4	25	AUSTIN	29060	525.94

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q6 Most Accessible

Database connection

Chicago Neighborhood Explorer

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Q6 Most Accessible

Rank the most transit-accessible neighborhoods.

Parameters

Number of neighborhoods to show

15

Run Query

Results

	community_id	community_area	num_stations	area_sq_miles	stations_per_sq_mile
0	32	LOOP	16	1.6621	9.63
1	37	FULLER PARK	2	0.7144	2.8
2	8	NEAR NORTH SIDE	7	2.7504	2.55
3	77	EDGEWATER	4	1.7379	2.3
4	38	GRAND BOULEVARD	4	1.7394	2.3
5	6	LAKE VIEW	7	3.1284	2.24
6	1	ROGERS PARK	4	1.8387	2.18
7	27	EAST GARFIELD PARK	4	1.9328	2.07

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q7 Crime Near Housing

Database connection

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Q7 Crime Near Housing

Compare crime patterns near affordable housing.

Parameters

Year

2025

Minimum housing units

10

Run Query

Results

	community_id	community_area	total_crimes	num_properties	affordable_units
0	25	AUSTIN	11788	12	703
1	8	NEAR NORTH SIDE	11138	24	1312
2	28	NEAR WEST SIDE	10310	35	1958
3	32	LOOP	8576	4	53
4	43	SOUTH SHORE	8109	6	412
5	24	WEST TOWN	6876	33	601
6	29	NORTH LAWNDALE	6420	35	1273
7	23	HUMBOLDT PARK	6152	48	1139

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q8 Transit Popularity

Database connection

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Q8 Transit Popularity

Rank neighborhoods by transit popularity.

Parameters

Number of neighborhoods to show

20

Year

2025

Run Query

Results

	station_id	station_name	community_area	total_entries	avg_weekday_rides
0	41660	Lake	LOOP	3595197	10549
1	40890	O'Hare	OHARE	3313419	9348
2	40380	Clark/Lake	LOOP	3202244	10293
3	40260	State/Lake	LOOP	2667263	7971
4	41450	Chicago	NEAR NORTH SIDE	2608031	7564
5	41220	Fullerton	LINCOLN PARK	2548433	7792
6	40370	Washington	LOOP	2531556	7749

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q9 Service Delays

Database connection

Q9 Service Delays

Find neighborhoods with the longest service delay patterns.

Parameters

Year

2025

Number of neighborhoods to show

20

Run Query

Results

	community_id	community_area	total_requests	avg_response_hours
0	9	EDISON PARK	3095	746.1
1	47	BURNSIDE	1174	611.71
2	45	AVALON PARK	4247	596.18
3	27	EAST GARFIELD PARK	8448	583.15
4	48	CALUMET HEIGHTS	6328	577.49
5	26	WEST GARFIELD PARK	7348	576.51
6	10	NORWOOD PARK	11674	574.32
7	29	NORTH LAWNDALE	12089	573.18
8	12	FOREST GLEN	6764	565.75

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q10 Demographics vs Crime

Database connection

Q10 Demographics vs Crime

Compare demographic indicators with crime levels.

Parameters

This query does not need any parameters.

Run Query

Results

	community_id	community_name	population	median_income	crime_rate
0	5	NORTH CENTER	35408	145156.86	35.16
1	12	FOREST GLEN	19517	138253.01	20.96
2	7	LINCOLN PARK	67831	137504.77	57.6
3	24	WEST TOWN	86598	134321.97	79.4
4	9	EDISON PARK	11203	133209.75	24.73
5	72	BEVERLY	19570	127056.19	50.33
6	33	NEAR SOUTH SIDE	29174	124967	83.4
7	8	NEAR NORTH SIDE	104712	122603.16	106.37
8	32	LOOP	42181	119836.79	203.31
9	74	MOUNT GREENWOOD	18553	115749.73	28.35

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Query Controls

Choose a query

Q11 Neighborhood Ranking

Database connection

Q11 Neighborhood Ranking

Show an overall ranked list of neighborhoods.

Parameters

Number of neighborhoods to show

20

Run Query

Results

	community_id	community_name	crime_rate	affordable_units	transit_density	avg_response_hours	livability_score
0	28	NEAR WEST SIDE	156.01	1958	1.76	13.1	0.629
1	38	GRAND BOULEVARD	129.09	2406	2.3	402.6	0.581
2	3	UPTOWN	72.54	1432	1.28	377	0.511
3	1	ROGERS PARK	69.08	1178	2.18	398.4	0.504
4	32	LOOP	203.31	53	9.63	278.8	0.504
5	33	NEAR SOUTH SIDE	83.4	1473	1.12	393.9	0.496
6	8	NEAR NORTH SIDE	106.37	1312	2.55	448.9	0.478
7	6	LAKE VIEW	59.41	523	2.24	353	0.461
8	77	EDGEWATER	53.57	529	2.3	425.5	0.444
9	34	ARMOUR SQUARE	81.4	147	2.01	271.9	0.425

Navigation

Choose a page

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Query Controls

Choose a query

Q12 Housing Availability

Database connection

Q12 Housing Availability

Summarize affordable housing availability.

Parameters

This query does not need any parameters.

Run Query

Results

	community_id	community_area	num_properties	total_units
0	38	GRAND BOULEVARD	30	2406
1	28	NEAR WEST SIDE	35	1958
2	33	NEAR SOUTH SIDE	12	1473
3	3	UPTOWN	39	1432
4	8	NEAR NORTH SIDE	24	1312
5	29	NORTH LAWNDALE	35	1273
6	27	EAST GARFIELD PARK	14	1205
7	1	ROGERS PARK	10	1178
8	23	HUMBOLDT PARK	48	1139
9	36	OAKLAND	18	1063
10	35	DOUGLAS	17	896

Navigation

Choose a page

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Query Controls

Choose a query

Q14 Ward Overlap

Database connection

Q14 Ward Overlap

Show which community areas overlap with a selected ward.

Parameters

Ward

27

Run Query

Results

	ward_id	community_id	community_name	overlap_sq_miles	pct_of_ward	pct_of_community_area	crime_rate	cost_burden_pct	avg_311_hours
0	27	28	NEAR WEST SIDE	2.1218	37.07	37.32	156.01	33.682	13.06
1	27	24	WEST TOWN	1.1399	19.91	24.91	79.4	27.892	374.36
2	27	23	HUMBOLDT PARK	1.0015	17.5	27.79	110.65	48.155	464.76
3	27	8	NEAR NORTH SIDE	0.8953	15.64	32.55	106.37	37.669	448.93
4	27	27	EAST GARFIELD PARK	0.5603	9.79	28.99	183.96	55.267	583.15
5	27	7	LINCOLN PARK	0.0051	0.09	0.16	57.6	29.285	333.56

Deploy

Navigation

Choose a page

- Run Queries
- Insert Service Request

Database connection

Chicago Neighborhood Explorer

Welcome! Use this dashboard to explore Chicago neighborhoods by housing, transit, crime, demographics, and service response patterns.

Insert Service Request

Request type

Street Light Out

Community area

ALBANY PARK

Street address

5801 S Ellis Ave

ZIP code

60637

Status

Open

Insert Service Request

Service request inserted successfully.